

Recombination shapes the diversification of the *wtf* meiotic drivers

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eLife Assessment

This **important** study provides one mechanism that can explain the rapid diversification of poison-antidote pairs in fission yeast: recombination between existing pairs. The evidence is largely **solid**, but the study can benefit from demonstrating that the novel poison-antidote constructed by the authors can serve as a meiotic driver. The work is of interest to colleagues studying genetic incompatibilities.

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Abstract

Meiotic drivers are selfish genetic elements that distort fair segregation. The *wtf* genes are poison-antidote meiotic drivers that are experiencing rapid diversification in fission yeasts. However, gene duplication alone is insufficient to drive the diversification of *wtf* genes, given the poison encoded by a newly duplicated *wtf* gene can be detoxified by the antidote encoded by the original *wtf* gene. Here, we analyze the evolution of *wtf* genes across 21 strains of *Schizosaccharomyces pombe*. Knocking out each of 25 *wtf* genes in *S. pombe* strain 972h separately does not attenuate the yeast growth, indicating that the *wtf* genes might be largely neutral to their carriers in asexual life cycle. Interestingly, *wtf* genes underwent recurrent and intricate recombination. As proof-of-principle, we generate a novel meiotic driver through artificial recombination between *wtf* drivers, and its encoded poison cannot be detoxified by the antidotes encoded by their parental *wtf* genes but can be detoxified by its own antidote. Therefore, we propose that recombination can generate new meiotic drivers and thus shape the diversification of the *wtf* drivers.

Introduction

During meiosis, the two alleles at a gene locus are separated into gametes, and each gamete has an equal chance of receiving either allele. This fundamental principle of inheritance, known as Mendel's law of segregation (Abbott and Fairbanks, 2016 ) , holds across most genetic loci in most sexual species. However, meiotic drivers, a class of selfish genetic elements, subvert fair

segregation during gametogenesis and are transmitted to more than one-half (even to all) of the functional gametes produced by a heterozygote (Sandler and Novitski, 1957 [↗](#); Lyttle, 1991 [↗](#); Hurst and Werren, 2002; Bravo Nunez et al., 2018b [↗](#)). Meiotic drivers can spread in a population even when they impose fitness costs on their hosts (Crow, 1991 [↗](#); Lindholm et al., 2016 [↗](#)). However, the spread of a meiotic driver can be thwarted by the costs imposed on its carriers or by its genetic suppressors (Lindholm et al., 2016 [↗](#)).

The fission yeast *wtf* (*with If Long Terminal Repeats*) gene family provides an excellent model to study how meiotic drivers act and evolve (Hu et al., 2017 [↗](#); Nuckolls et al., 2017 [↗](#)). Many *wtf* genes are autonomous one-gene poison-antidote meiotic drivers that encode both a spore-killing poison (short isoform) and an antidote to the poison (long isoform) using alternative transcriptional initiation (Hu et al., 2017 [↗](#); Nuckolls et al., 2017 [↗](#); Nuckolls et al., 2022). To achieve meiotic drive, all spores are exposed to the poison, whereas only those that inherit *wtf* express the antidote and are rescued (Hu et al., 2017 [↗](#); Nuckolls et al., 2017 [↗](#); Nuckolls et al., 2022). Some other *wtf* genes can act as drive suppressors (Bravo Nunez et al., 2018a [↗](#); Bravo Nunez et al., 2020a [↗](#)). The poison and the antidote differ only in their N-terminal cytosolic tails containing PY (Leu/Pro-Pro-X-Tyr) motifs. PY motif-dependent ubiquitination promotes the transport of the antidote and the poison (physically interacted with the antidote) from the trans-Golgi network to the endosome, thereby preventing toxicity (Zheng et al., 2023 [↗](#)).

The *wtf* gene family is experiencing rapid diversification: the *Schizosaccharomyces pombe* reference genome encodes 25 *wtf* genes, some of which are pseudogenes. The copy numbers of *wtf* genes vary greatly among different *S. pombe* strains, and frequent nonallelic gene conversion occurs between *wtf* genes (Hu et al., 2017 [↗](#); Nuckolls et al., 2017 [↗](#); Eickbush et al., 2019 [↗](#)). However, these findings are based on a limited number of strains, and the pattern and extent of recombination in the *wtf* genes remain to be fully explored. Moreover, *wtf* driver genes are present in the last common ancestor (LCA) of the fission yeasts *S. pombe*, *S. octosporus*, *S. osmophilus*, and *S. cryophilus*, indicating that *wtf* genes have likely maintained the capacity to drive for more than 100 million years (De Carvalho et al., 2022 [↗](#)). These fission yeast species carry varying numbers of *wtf* genes, ranging from 5 to 83 (De Carvalho et al., 2022 [↗](#)). Yet, it remains perplexing how *wtf* genes achieved such diversification. On one hand, gene duplication can give birth to new *wtf* gene copies. On the other hand, a newly duplicated *wtf* gene might not drive because the poison produced by the newly duplicated *wtf* gene can be detoxified by the original *wtf* gene. Like newly duplicated genes, the most probable fate of a new *wtf* duplicate is pseudogenization (Lynch, 2007 [↗](#); Innan and Kondrashov, 2010 [↗](#)). Thus, the vast majority of new *wtf* duplicates experience an early exit from the population, most probably never reaching fixation (Lynch, 2007 [↗](#); Innan and Kondrashov, 2010 [↗](#)). To become a new driver, the new *wtf* copy should evolve coupling new poison and new antidote to the new poison through mutations. But the fate-changing mutations are likely to be rare. It follows that gene duplication might be insufficient to drive the diversification of *wtf* genes.

In this study, we analyzed the diversity and evolution of *wtf* genes in the genomes of 21 strains of *S. pombe* that were sequenced using long-read sequencing approaches (Tusso et al., 2022 [↗](#)). Through knocking out each of 25 *wtf* genes in *S. pombe* laboratory strain 972h-, no significant attenuated growth was observed, indicating *wtf* genes might be not deleterious in the asexual life cycle. We found that recurrent recombination occurred among *wtf* genes. We generated a novel meiotic driver through artificial recombination between *wtf* drivers, and its encoded poison cannot be detoxified by the antidotes encoded by their parental *wtf* genes but can be detoxified by its own antidote. Therefore, we propose that recombination can generate *wtf* driver with new poisons and might shape the diversification of *wtf* genes.

Results

Diversity and evolution of *wtf* genes in fission yeasts

First, we analyzed the diversity and distribution of *wtf* genes in fission yeasts. The *S. pombe* reference genome (strain 972h-) encodes a total of 25 *wtf* genes. For these 25 *wtf* genes, the number of exons varies from 3 to 6 (**Figure 1A**) (Bowen et al., 2003). To investigate the relationship among exons from different *wtf* genes, we grouped these *wtf* exons into clusters based on nucleotide identity of 0.50 (**Figure 1B** and **C**). The *wtf* exons were grouped into 10 clusters with >2 members, and 12 exons exist as singletons in the similarity network (**Figure 1B** and **C**; Supplementary file 1c). Only exon 1 of all the 25 *wtf* genes group together in a cluster, indicating that the first exons are well conserved among the *wtf* genes. No other exon is conserved among all the 25 *wtf* genes. Therefore, the evolution of the *wtf* gene structures appear to be highly dynamic.

We next identified *wtf* genes in 21 strains of *S. pombe* that were sequenced using long-read sequencing approaches (Supplementary file 1b) (Tusso et al., 2022). The copy number of *wtf* genes varies among different *S. pombe* strains, ranging from 24 (strain JB879) to 37 (strain JB1206) (**Figure 1D**). Synteny analyses show that the *wtf* genes are present in 20 genetic loci (**Figure 1D**). Multiple *wtf* genes were present in 13 *wtf* loci. Within 20 *wtf* loci, at least one *wtf* gene is present in all of or nearly all of the 21 *S. pombe* strains, suggesting that these 20 *wtf* loci might have originated before the LCA of the 21 *S. pombe* strains. Putative *wtf* pseudogenes are prevalent in many *wtf* loci among the 21 *S. pombe* strains, indicating frequent pseudogenization occurred in the *wtf* genes. These results indicate that *wtf* copy number variation is prevalent among *S. pombe* strains.

We performed phylogenetic analyses of the *wtf* genes from 21 *S. pombe* strains and three other fission yeast species (*S. octosporus*, *S. cryophilus*, and *S. osmophilus*) (**Figure 1E**; Supplementary file 1d). The *wtf* genes of *S. pombe* form a monophyletic group. Orthologs of *wtf14*, *wtf7*, *wtf11*, and *wtf15* form monophyletic groups, whereas orthologs of other *wtf* genes show complex phylogenetic mixing, indicating complex recombination might have occurred among these *wtf* genes (**Figure 1E**) (Eickbush et al., 2019). Moreover, the *wtf* genes with 6 exons (including the known functional drivers *wtf4* and *wtf23*) cluster together, and exhibit a ladder-like phylogeny, which might be generated by continual selection driven by antidotes (like the ladder-like phylogeny of influenza A viruses H1N1 and H3N2, which is shaped by continual immune selection (Grenfell et al., 2004; Bedford et al., 2011)). Based on phylogenetic relationship, we divided the *wtf* genes of 21 *S. pombe* strains into eight groups, namely groups 1 to 8, among which groups 5 to 8 include orthologs of *wtf14*, *wtf7*, *wtf11*, and *wtf15*, respectively (**Figure 1E**). Exon 2 ATG codons (exon-2-ATG) and in-frame ATG within intron 1 and near the start of exon 2 (intron-1-ATG) of *wtf* genes can encode the start of poison protein isoforms (Hu et al., 2017). We found that most of exon-2-ATG and intron-1-ATG are present within group 1 *wtf* genes (**Figure 1E**). A majority of the *wtf* genes are flanked by solo-LTRs (Bowen et al., 2003) (**Figure 1E**). However, the solo-LTRs flanking the *wtf* genes do not cluster together but form many distinct groups, suggesting that solo-LTRs were inserted nearby the *wtf* genes multiple times (Supplementary file 1a). Together, our results reveal the rapid diversification and turnover of *wtf* genes in a single fission yeast species.

No attenuated growth of fission yeast without *wtf* genes

To explore the effect of *wtf* genes on the fitness of fission yeast, we knocked out each of the 25 *wtf* genes in the *S. pombe* laboratory strain 972h-using a method based on homologous recombination (**Figure 2A**). A total of 25 *wtf* knockout strains ($\Delta wtf1$ to $\Delta wtf25$) were generated. We used spot assay to evaluate the effect of *wtf* gene knockout on the yeast growth, and no growth defect was

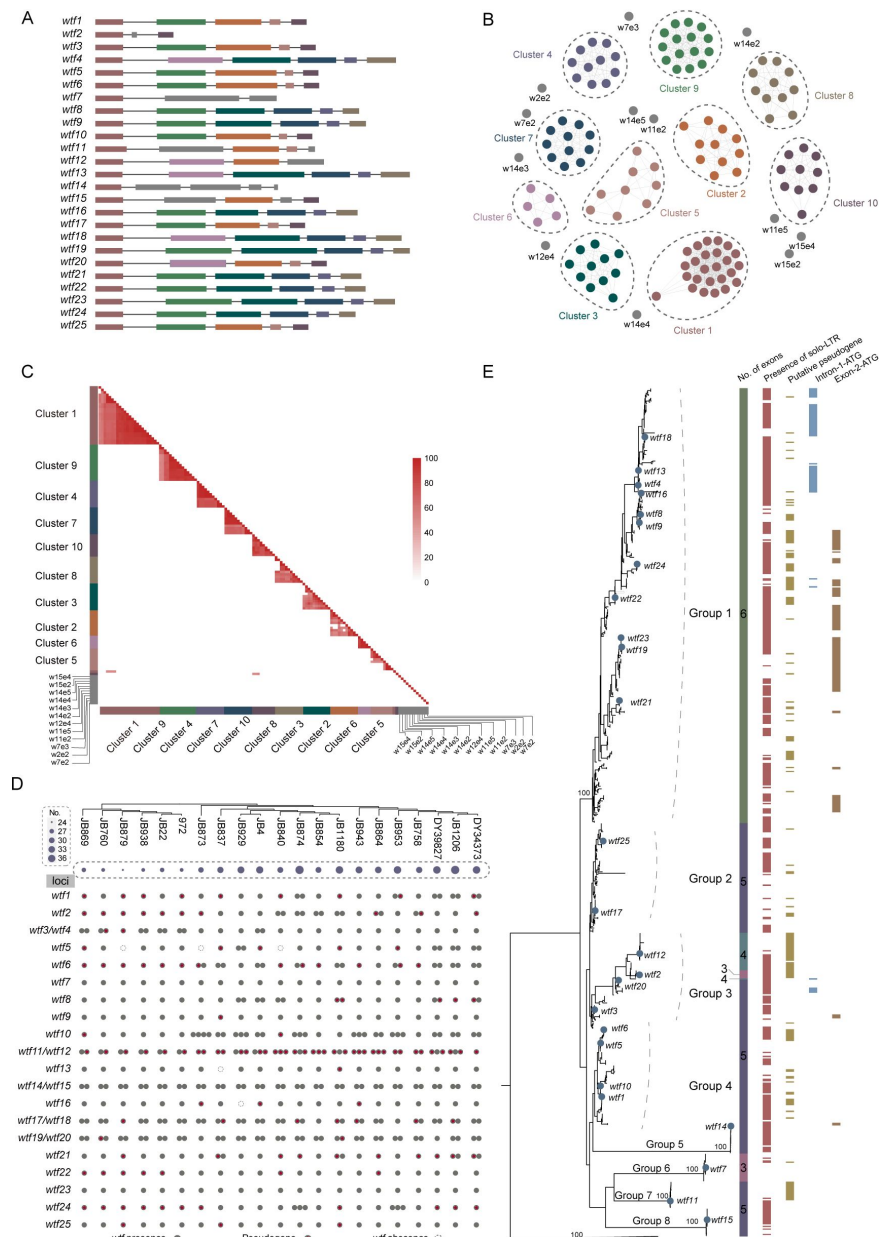


Figure 1.

The diversity and evolution of *wtf* genes in fission yeasts.

(A) The gene structures of the *wtf* genes in the *S. pombe* reference genome. Rectangles and lines represent exons and introns, respectively. Rectangles with the same color indicate exons that share >50% sequence identity. (B) The similarity network of exons of the *wtf* genes in the *S. pombe* reference genome. Sequences that share an identity of >50% form a cluster. The colors of *wtf* exons correspond to these in panel (A). The exons for each cluster were listed in the Supplementary file 1c. (C) The heatmap of nucleotide identity among exons of the *wtf* genes in *S. pombe* reference genome. (D) The distribution *wtf* genes in 21 *S. pombe* strains. *wtf* genes were present in 25 genetic loci. Filled circles and empty circles represent the presence or absence of *wtf* genes, respectively. Red pentagons indicate putative *wtf* pseudogenes. The size of circles in purple indicates the number of *wtf* genes. The relationship among 21 *S. pombe* strains was inferred based on phylogenetic analysis of 30 randomly selected genes. (E) Phylogenetic relationship of *wtf* genes in 21 *S. pombe* strains. The phylogenetic tree is reconstructed using the maximum likelihood method. Filled circles in blue indicate *wtf* genes in *S. pombe* reference genome. The number of exons, the presence of solo-LTRs, the putative pseudogene status, the presence of intron-1-ATG codon, the presence of exon-2-ATG codon is shown near the corresponding *wtf* gene. *wtf* genes from other fission yeast species were collapsed into a triangle.

observed for all the 25 *wtf* knockout strains (**Figure 2B**). Therefore, our experiment suggests that the *wtf* genes might be largely neutral to the fitness of their carriers in the asexual life cycle at least in normal growth condition.

Recurrent recombination in *wtf* genes

Given complex phylogenetic mixing observed among *wtf* genes (**Figure 1E**), we tested whether recombination occurred. We detected signals of recombination in the 25 *wtf* genes of the *S. pombe* reference genome ($p = 0$) and in the *wtf* genes of the 21 *S. pombe* strains ($p = 0$) using pairwise homoplasy index (HPI) test. Split network analysis also supports the frequent occurrence of recombination in the 25 *wtf* genes of the *S. pombe* reference genome (**Figure 3A**) and in the *wtf* genes of the 21 strains of *S. pombe* (**Figure 3B**). In contrast, no recombination signal was tested for groups 5 to 8 using HPI test ($p = 1$ for group 5, $p = 1$ for group 6, $p = 0.53$ for group 7, and $p = 1$ for group 8). We estimated recombination rates of the full-length *wtf* sequences, the first exons, and the *wtf* sequences without the first exons for *wtf* groups 1 to 4. We found that the recombination rate of group 1 *wtf* was highest among the four *wtf* groups (**Figure 3C**). For group 1, breakpoints are dispersed across the *wtf* sequences (**Figure 3D**). These lines of evidence suggest that *wtf* genes underwent recurrent and intricate recombination.

Generation of a new driver gene through artificial recombination

Given gene duplication alone might be insufficient to shape the diversification of *wtf* genes, we hypothesize that recombination between *wtf* genes can generate new meiotic drives. To test this, we constructed four chimeric *wtf* genes through recombination among known functional meiotic drivers (*wtf23*) and an artificially generated meiotic driver (*wtf18*).

We used a proved *Saccharomyces cerevisiae* system to test the activity of poison and antidote proteins encoded by *wtf* genes (Nuckolls et al., 2020). As expected, the expression of the poison proteins (Wtf23^{poison}) encoded by *wtf23* genes caused the yeast growth arrest (**Figure 4A**). The attenuated growth was alleviated, when the corresponding antidote proteins (Wtf23^{antidote}) were expressed (**Figure 4A**). We also experimentally analyzed *wtf18* gene, which was known to encode only long (antidote-like) transcripts and probably act as a suppressor (Bravo Nunez et al., 2018a). We artificially introduced an in-frame ATG codon right before the start of exon 2, generating *wtf18*^{poison/-OM}. The expression of *wtf18*^{poison/-OM} resulted in the yeast growth arrest, suggesting its product, Wtf18^{poison/-OM}, is indeed a poison protein (**Figure 4B**). When co-expressing *wtf18*^{antidote} and *wtf18*^{poison/-OM}, the attenuated yeast growth was rescued (**Figure 4B**), indicating that Wtf18^{antidote} can ameliorate the toxicity of Wtf18^{poison/-OM}. Therefore, the introduction of an ATG codon to *wtf18* leads to the generation of a putative meiotic driver.

We then constructed four chimeric *wtf* genes through artificial recombination between *wtf23* and *wtf18*, including *wtfC1* (possessing exons 1-2 of *wtf23* and exons 3-6 of *wtf18*), *wtfC2* (possessing exons 1-3 of *wtf23* and exons 4-6 of *wtf18*), *wtfC3* (possessing exons 1-4 of *wtf23* and exons 5-6 of *wtf18*), and *wtfC4* (possessing exons 1-5 of *wtf23* and exon 6 of *wtf18*). The expression of the short isoforms of *wtfC1*, *wtfC2*, *wtfC3*, and *wtfC4* resulted in yeast growth arrest, revealing their toxicity (**Figure 4C-F**). However, the antidote of *wtfC1* and *wtfC3* cannot detoxify the corresponding chimeric toxins (**Figure 4C** and **E**). Interestingly, we generated a putative novel meiotic driver, namely *wtfC4*. Our results show that *wtfC4* encodes a functional poison (WtfC4^{poison}) (**Figure 4F**). The poison can be detoxified by its own long isoforms (dubbed as *wtfC4*^{antidote}), but cannot be detoxified by the antidote proteins of their parental genes (**Figure 4F**). Taken together, we generated a new meiotic driver through artificial recombination between pre-existing *wtf* genes.

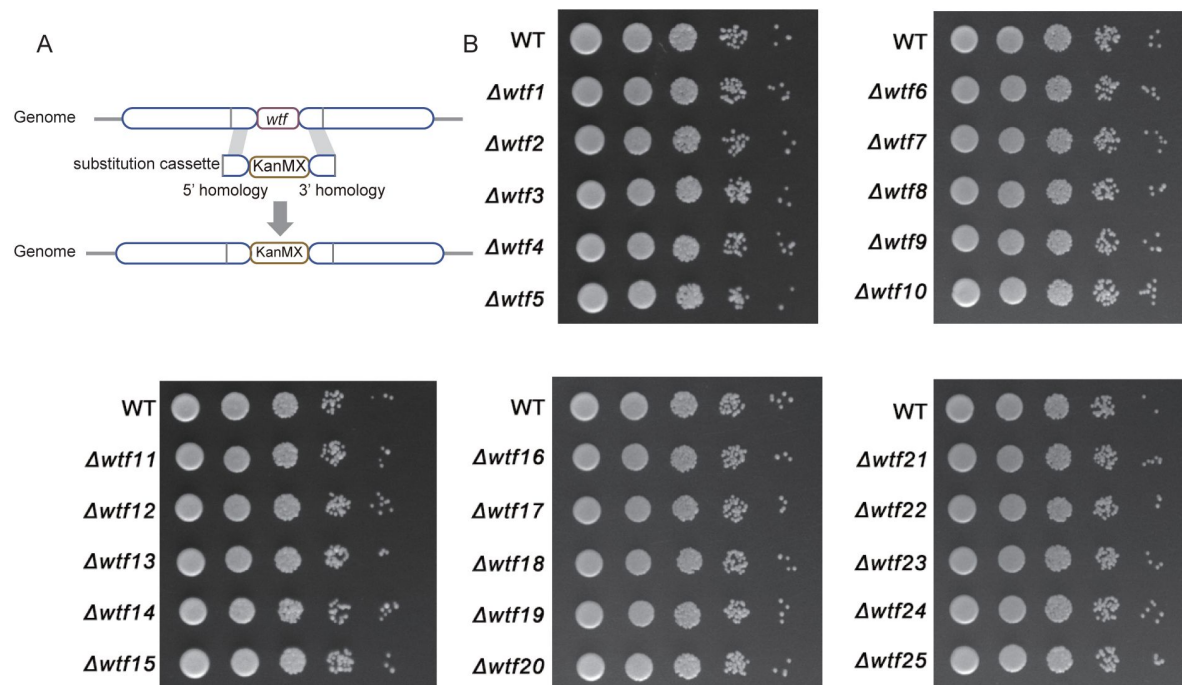


Figure 2.

The effect of *wtf* gene on the growth of *S. pombe*.

(**A**) Generation of *wtf* knockout (Δwtf) strains based on the homologous recombination method. Substitution cassette contains a kanMX resistance marker and two homologous sequences flanking the target *wtf* genes. (**B**) Spot assay of Δwtf strains. The strains were continuously diluted by a 10-fold gradient to 10^{-5} , and five stock suspensions were spotted onto the YE (Yeast extract) liquid medium.

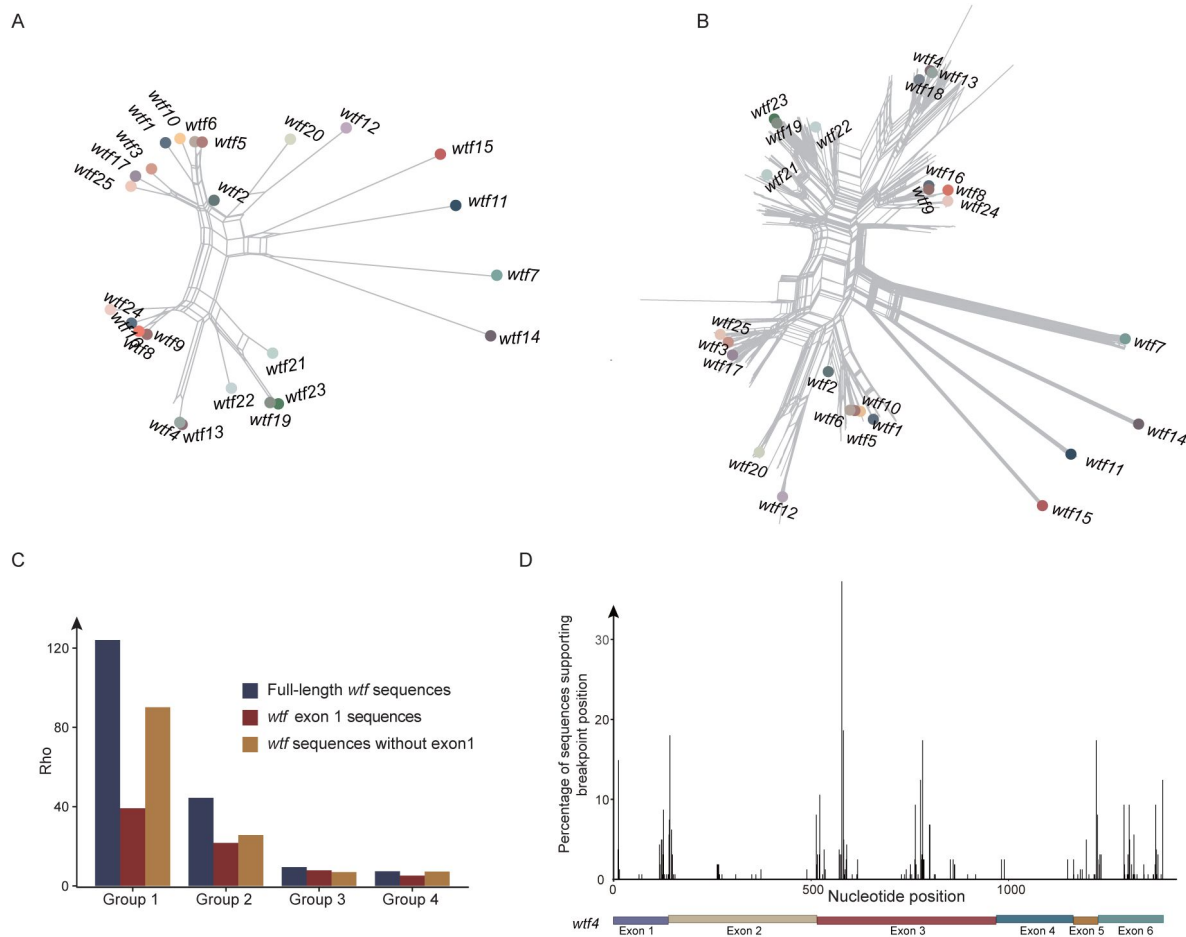


Figure 3.

Recombination analysis of *wtf* genes.

(A) Split tree of 25 *wtf* genes of the *S. pombe* reference genome. (B) Split tree of *wtf* genes from 21 *S. pombe* strains. The *wtf* genes of the *S. pombe* reference genome are labelled. (C) Recombination rates of *wtf* genes that belong to groups 1 to 4. Recombination rates were estimated for the full-length *wtf* sequences, the first exons, and the *wtf* sequences without the first exons. (D) Breakpoints detected for *wtf* genes of group1. The exons of *wtf4*, as a gene position reference, are shown.

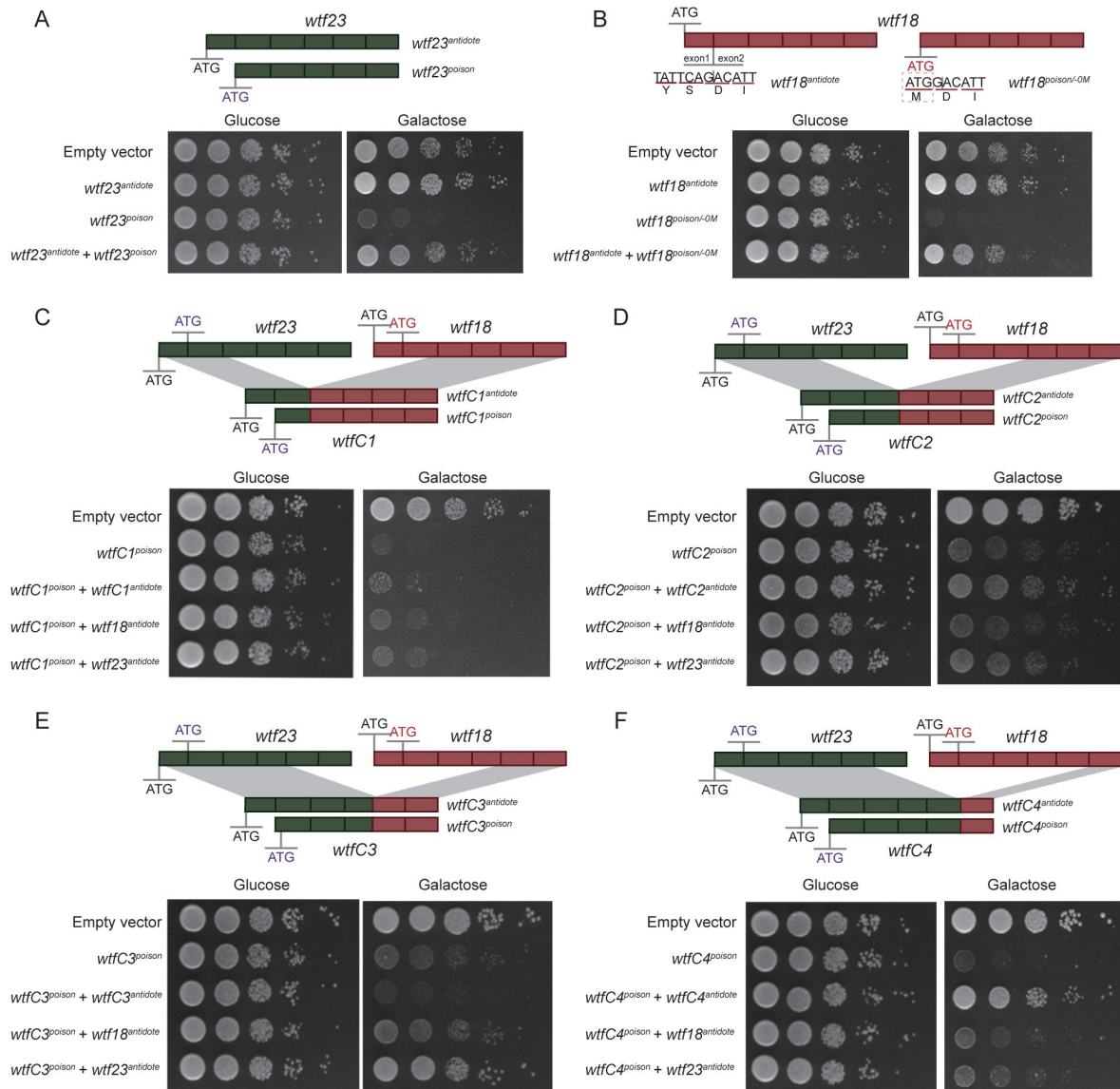


Figure 4.

Poison and antidote activity of *wtf* genes and chimeric *wtf* genes.

wtf23 and *wtf18* are highlighted in green and red, respectively. Rectangles represent exons. The start codon ATG is shown. For *wtf18*, an in-frame ATG codon was introduced right before the start of exon 2, generating *wtf18^{poison/-OM}*. Spot assay of yeast transformed with the short isoforms (encoding poison-like proteins) and the long isoforms (encoding antidote-like proteins) of *wtf23* (A) and *wtf18* (B). (C to F) Four chimeric *wtf* genes were generated through artificial recombination, namely *wtfC1* (with exons 1-2 of *wtf23* and exons 3-6 of *wtf18*), *wtfC2* (with exons 1-3 of *wtf23* and exons 4-6 of *wtf18*), *wtfC3* (with exons 1-4 of *wtf23* and exons 5-6 of *wtf18*), and *wtfC4* (with exons 1-5 of *wtf23* and exon 6 of *wtf18*). Spot assay of yeast transformed with the short isoforms (encoding poison-like proteins) and the long isoforms (encoding antidote-like proteins) of *wtfC1* (C), *wtfC2* (D), *wtfC3* (E), and *wtfC4* (F) are shown.

Discussion

In this study, we analyzed the diversity and evolution of *wtf* genes in fission yeasts. The copy number of the *wtf* gene varies among different *S. pombe* strains, revealing rapid diversification and turnover of the *wtf* genes within a single fission yeast species. We detected signals of recurrent and intricate recombination among *wtf* genes as previously reports with limited genomes (Hu et al., 2017 [↗](#); Nuckolls et al., 2017 [↗](#); Eickbush et al., 2019 [↗](#); De Carvalho et al., 2022 [↗](#)). We hypothesize that recombination between *wtf* genes can produce new *wtf* genes with new poisons and the antidotes to new poisons. These new *wtf* genes can then driver through populations. As proof-of-principle, we generated a chimeric *wtf* gene that represents a new meiotic driver. The encoded poison of the newly generated meiotic driver can be detoxified by its own long isoforms, but cannot be detoxified by the antidote proteins of their parental genes. However, the other three chimeric *wtf* genes did not show this property, possibly because the last exon might be crucial for antidote function. Indeed, our recombination breakpoint analyses (**Figure 3D** [↗](#)) reveal substantial recombination might have occurred in the last exon. Together, our results indicate that recombination is likely to drive the rapid diversification of *wtf* gene in fission yeasts.

Most of the known meiotic drivers impose costs on their carriers due to direct effects of the driver on survival or fertility, production of a biased sex ratio, or via deleterious mutations linked to the driver (Price and Wedell, 2008 [↗](#); Larracuenta et al., 2012; Sutter and Lindholm, 2015 [↗](#); Fishman et al., 2015; Lindholm et al., 2016 [↗](#); Zanders and Unckless, 2019 [↗](#)). In outcrossing between individuals from distinct yeast lineages, *wtf* drivers can provide a selective advantage to atypical spores, such as aneuploids and diploids (Bravo Nunez et al., 2020b [↗](#)). In this study, we assessed the effects of the *wtf* genes on the growth of fission yeast during the asexual life cycle through knocking out each of the 25 *wtf* genes in *S. pombe* laboratory strain 972h-separately. We did not observe obvious attenuated growth for these *wtf* knockout strains, indicating *wtf* genes are largely neutral to the fitness of their carriers during the asexual life cycle at least in the normal growth setting. It should be noted that the spot assay used in this study detect large differences in fitness between wide type and *wtf* knockout strains. Nevertheless, it is likely that *wtf* genes evolve mainly in a neutral manner during the asexual life cycle, which explains the presence of high proportion of pseudogenes in *wtf* gene repertoire. Moreover, asexual reproduction is much more frequent than sexual reproduction for yeasts (Tsai et al., 2008). Therefore, even if fate-changing mutations that simultaneously produce new poison and the antidote to new poison occur, the most probable fate of a new *wtf* gene generated by gene duplication is pseudogenization and removed from the population.

Gene duplication gives rise to new *wtf* genes. However, the newly generated *wtf* gene can be detoxified by the original *wtf* gene and thus cannot drive through its host population, when the original *wtf* is fixed in the population (**Figure 5** [↗](#)). Therefore, most, if not all, of *wtf* gene duplicates experience early exist from the host population. When recombination occurs between two pre-existing *wtf* genes, chimeric *wtf* gene with new poison and the antidote to new poison can be generated as this study shows. Then, the *wtf* gene with new driver property can spread in its host population, even reaching fixation (**Figure 5** [↗](#)). During asexual life cycle, *wtf* genes evolve mainly under genetic drift, and thus can accumulate disruptive mutations, leading to their pseudogenization. Taken together, our study highlights the significance of recombination in shaping the diversification of *wtf* genes.

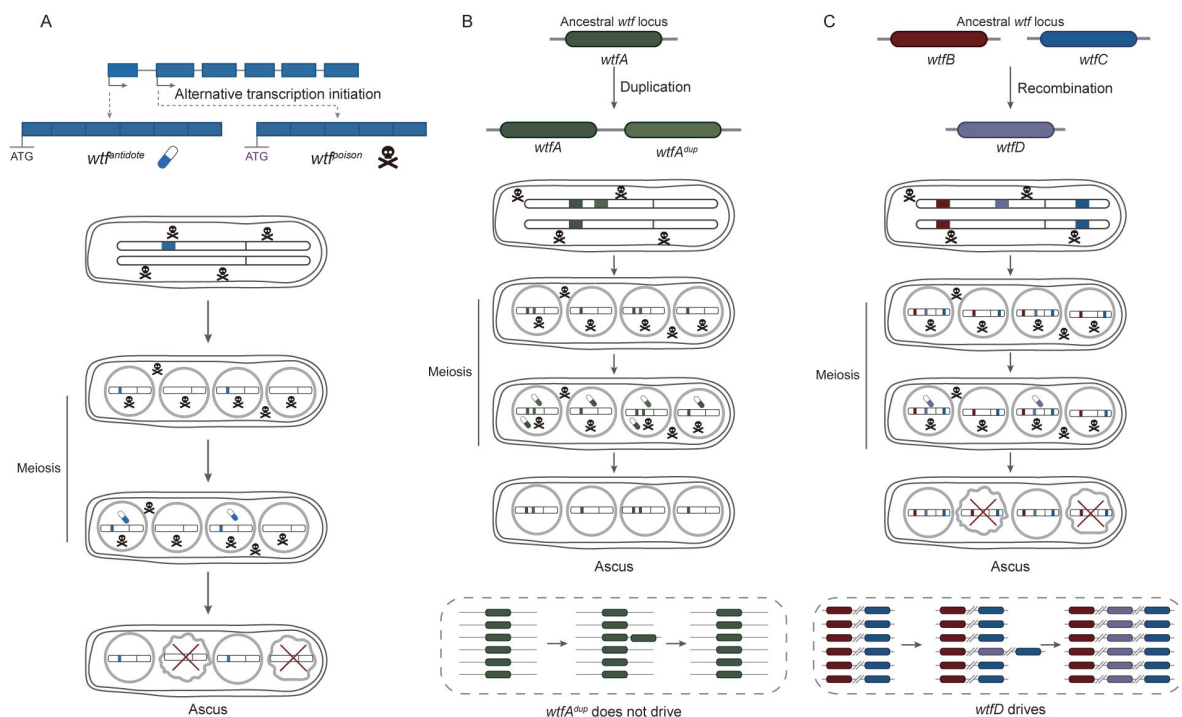


Figure 5.

The evolutionary mechanisms of *wtf* genes.

(A) Mechanism of *wtf* meiotic driver. *wtf* genes are transcribed into two transcripts, namely long and short isoforms. The long and short isoforms encode antidote and poison, respectively. All spores are exposed to the poison, whereas only those that inherit *wtf* genes express the antidote and are rescued. Spores without the *wtf* allele are destroyed, resulting in the drive of *wtf*. **(B)** Duplication is insufficient to drive the diversification of *wtf* genes. Gene duplication of *wtfA* gives rise to a new *wtf* gene, *wtfA^{dup}*. However, *wtfA^{dup}* gene can be detoxified by the original *wtfA* gene and thus cannot drive through its host population. **(C)** When recombination occurs between two parental *wtfB* and *wtfC* genes, generating chimeric *wtfD* gene. *wtfD* encodes new poison that cannot be detoxified by their parental antidote and the antidote to new poison. The *wtfD* gene then can spread in its host population.

Methods

Identification of the *wtf* genes

We used the *blastn* algorithm to identify *wtf* genes within 21 *S. pombe* strains with 25 *wtf* genes from *S. pombe* reference genome as the queries and an *e*-cutoff value of 10^{-5} . The identified *wtf* genes were annotated based on the *wtf* genes of reference genome. The sequence identity among the exons of *wtf* genes was calculated using BioAider version 1.334 (Zhou et al., 2020). Exons was then clustered based on the nucleotide identity using igraph package version 2.0.1.1 (Csardi et al., 2006; Csardi et al., 2024). We extended 1000 bp flanking each *wtf* gene to establish their syntenic relationships.

Phylogenetic analysis

The coding sequences of *wtf* genes of 21 *S. pombe* strains and three other fission yeast species (*S. octosporus*, *S. cryophilus*, and *S. osmophilus*) were aligned using MAFFT version 7 (Katoh and Standley, 2013). To clarify the relationship of 21 *S. pombe* strains, 30 genes were randomly selected and concatenated using Phylosuite version 1.2.1 (Zhang et al., 2020). All the phylogenetic analyses in this study were performed using the maximum likelihood (ML) method implemented in IQ-TREE version 2 (Minh et al., 2020). The best-fit substitution model was selected using the ModelFinder algorithm (Kalyaanamoorthy et al., 2017). Node supports were assessed using the ultrafast bootstrap approximation (UFBoot) method with 1,000 replicates (Hoang et al., 2018). Solo-LTRs were identified using the *blast* algorithm and aligned using MAFFT version 7 (Katoh and Standley, 2013). Phylogenetic analysis was performed using the approximate maximum likelihood method implemented in FastTree version 2.1.1 (Price et al., 2010).

Recombination analysis

Split networks of *wtf* genes were generated using the neighborhood network analysis implemented in Splittree4 (Huson et al., 2006). Pairwise homoplasy index (PHI) test was performed using SplitsTree4 (Huson et al., 2006). Potential breakpoints were detected using 3SEQ (Lam et al., 2018). The recombination rate was estimated using FastEPRR package version 2.0 (Gao et al., 2016).

Generation of *wtf* knockout strains

The *wtf* knockout (Δwtf) strains generated in this study were derived from *S. pombe* strain 972h-. We constructed substitution cassettes for each of the 25 *wtf* genes of the *S. pombe* reference genome. Substitution cassettes contain a kanMX resistance marker and two homologous sequences flanking the target *wtf* genes (Moreno et al., 1991; García-Ríos et al., 2014). Substitution cassettes were transformation into *S. pombe* (strain 972h-) through the lithium acetate-based method (Moreno et al., 1991; García-Ríos et al., 2014). *wtf* gene knockout strains were selected for kanMX resistance and were verified by PCR.

Plasmid construction

Total RNA of fission yeast was extracted and reverse transcribed into cDNA. Coding sequences of *wtf23^{antidote}*, *wtf23^{poison}*, and *wtf18^{antidote}* were amplified using the corresponding primers (Supplementary file 1e). *wtf18^{poison/-M0}* was generated using *wtf18^{antidote}* as the template and the primer with an artificially introduced ATG (Supplementary file 1e). We generated *wtfC1* through recombining exons 1-2 of *wtf23* and exons 3-6 of *wtf18*, generated *wtfC2* through recombining exons 1-3 of *wtf23* and exons 4-6 of *wtf18*, generated *wtfC3* through recombining exons 1-4 of *wtf23* and exons 5-6 of *wtf18*, and generated *wtfC4* through recombining exons 1-5 of *wtf23* and exon 6 of

wtf18. These *wtf* and *wtfC* genes were then cloned into the GAL1/10 dual expression plasmid Gal_HF. Plasmids were first transformed into *Escherichia coli* and verified by PCR and sequencing. Plasmids were then transformed into *S. cerevisiae* (strain S288C) using the lithium acetate-based method. Yeast transformants were selected for kanMX resistance and were verified by PCR.

Spot assay

The yeast strains were cultured in YPD liquid medium at 30°C with shaking at 200 rpm. The overnight cultures were transferred to fresh YPD liquid medium and grown to an OD₆₀₀ value of ~3. Cells were collected by centrifugation, and the OD₆₀₀ was adjusted to 3. Subsequently, the strains were continuously diluted by a 10-fold gradient to 10⁻⁵, generating five stock suspensions. The five stock suspensions were used for spot assays. We plated 1.5 µL of each stock suspension on the surface of YPD and YPG solid media. The plates were incubated at 30°C, and the growth of colonies were observed.

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Data Availability

No new data were generated in support of this research.

References

- Abbott S, Fairbanks DJ (2016) **Experiments on Plant Hybrids by Gregor Mendel** *Genetics* **204**:407–422
- Bedford T, Cobey S, Pascual M (2011) **Strength and tempo of selection revealed in viral gene genealogies** *BMC Evol Biol* **11**
- Bowen NJ, Jordan IK, Epstein JA, Wood V, Levin HL (2003) **Retrotransposons and their recognition of pol II promoters: a comprehensive survey of the transposable elements from the complete genome sequence of *Schizosaccharomyces pombe*** *Genome Res* **13**:1984–1997
- Bravo Nunez MA, Lange JJ, Zanders SE (2018) **A suppressor of a wtf poison-antidote meiotic driver acts via mimicry of the driver's antidote** *PLoS Genet* **14**
- Bravo Nunez MA, Nuckolls NL, Zanders SE (2018) **Genetic Villains: Killer Meiotic Drivers** *Trends Genet* **34**:424–433
- Bravo Nunez MA, Sabbarini IM, Eickbush MT, Liang Y, Lange JJ, Kent AM, Zanders SE (2020) **Dramatically diverse *Schizosaccharomyces pombe* wtf meiotic drivers all display high gamete-killing efficiency** *PLoS Genet* **16**
- Bravo Nunez MA, Sabbarini IM, Eide LE, Unckless RL, Zanders SE (2020) **Atypical meiosis can be adaptive in outcrossed *Schizosaccharomyces pombe* due to wtf meiotic drivers** *eLife* **9**
- Crow JF (1991) **Why is Mendelian so exact?** *Bioessays* **13**:305–12
- Csardi G, Nepusz T (2006) **The igraph software package for complex network research."** *InterJournal Complex Systems* **1695**
- Csárdi G, Nepusz T, Traag V, Sz Horvát, Zanini F, Noom D, Müller K (2024) **igraph: Network Analysis and Visualization in R** *CRAN*
- De Carvalho M, Jia GS, Nidamangala Srinivasa A, Billmyre RB, Xu YH, Lange JJ, Sabbarini IM, Du LL, Zanders SE (2022) **The wtf meiotic driver gene family has unexpectedly persisted for over 100 million years** *Elife* **11**
- Eickbush MT, Young JM, Zanders SE (2019) **Killer Meiotic Drive and Dynamic Evolution of the wtf Gene Family** *Mol Biol Evol* **36**:1201–1214
- Fishman L, Kelly JK (2015) **Centromere-associated meiotic drive and female fitness variation in *Mimulus*** *Evolution* **69**:1208–18
- Gao F, Ming C, Hu W, Li H (2016) **New Software for the Fast Estimation of Population Recombination Rates (FastEPRR) in the Genomic Era** *G3* **6**:1563–71

- García-Ríos E, Gutiérrez A, Salvadó Z, Arroyo-López FN, Guillaumon JM (2014) **The fitness advantage of commercial wine yeasts in relation to the nitrogen concentration, temperature, and ethanol content under microvinification conditions** *Appl Environ Microbiol* **80**:704–13
- Grenfell BT, Pybus OG, Gog JR, Wood JL, Daly JM, Mumford JA, Holmes EC (2004) **Unifying the epidemiological and evolutionary dynamics of pathogens** *Science* **303**:327–32
- Hoang DT, Chernomor O, von Haeseler A, Minh BQ, Vinh LS (2018) **UFBoot2: Improving the Ultrafast Bootstrap Approximation** *Mol Biol Evol* **35**:518–522
- Huson DH, Bryant D (2006) **Application of Phylogenetic Networks in Evolutionary Studies** *Mol Biol Evol* **23**:254–267
- Hu W, Jiang ZD, Suo F, Zheng JX, He WZ, Du LL (2017) **A large gene family in fission yeast encodes spore killers that subvert Mendel's law** *Elife* **6**
- Hurst GD, Werren JH (2001) **The role of selfish genetic elements in eukaryotic evolution** *Nat Rev Genet* **2**:597–606
- Innan H, Kondrashov F (2010) **The evolution of gene duplications: classifying and distinguishing between models** *Nat Rev Genet* **11**:97–108
- Kalyaanamoorthy S, Minh BQ, Wong TKF, von Haeseler A, Jermini LS (2017) **ModelFinder: fast model selection for accurate phylogenetic estimates** *Nat Methods* **14**:587–589
- Katoh K, Standley DM (2013) **MAFFT multiple sequence alignment software version 7: improvements in performance and usability** *Mol Biol Evol* **30**:772–80
- Lam HM, Ratmann O, Boni MF (2018) **Improved algorithmic complexity for the 3SEQ recombination detection algorithm** *Mol Biol Evol* **35**:247–251
- Larracuent AM, Presgraves DC (2012) **The selfish Segregation Distorter gene complex of *Drosophila melanogaster*** *Genetic* **192**:33–53
- Lindholm AK *et al.* (2016) **The Ecology and Evolutionary Dynamics of Meiotic Drive** *Trends Ecol Evol* **31**:315–326
- Lyttle TW (1991) **Segregation distorters** *Annu Rev Genet* **25**:511–57
- Lynch M (2007) **The Origins of Genome Architecture** Sunderland (MA): Sinauer Associates
- Moreno S, Klar A, Nurse P (1991) **Molecular genetic analysis of fission yeast *Schizosaccharomyces pombe*** *Methods Enzymol* **194**:795–823
- Minh BQ, Schmidt HA, Chernomor O, Schrempf D, Woodhams MD, von Haeseler A, Lanfear R (2020) **IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era** *Mol Biol Evol* **37**:1530–1534
- Nuckolls NL, Bravo Núñez MA, Eickbush MT, Young JM, Lange JJ, Yu JS, Smith GR, Jaspersen SL, Malik HS, Zanders SE (2017) **wtf genes are prolific dual poison-antidote meiotic drivers** *eLife* **6**

Nuckolls NL *et al.* (2020) **The wtf4 meiotic driver utilizes controlled protein aggregation to generate selective cell death** *Elife* **9**

Nuckolls NL, Nidamangala Srinivasa A, Mok AC, Helston RM, Bravo Núñez MA, Lange JJ, Gallagher TJ, Seidel CW, Zanders SE (2020) **S. pombe wtf drivers use dual transcriptional regulation and selective protein exclusion from spores to cause meiotic drive** *PLoS Genet* **18**

Price MN, Dehal PS, Arkin AP (2010) **FastTree 2--approximately maximum-likelihood trees for large alignments** *PLoS One* **5**

Price TA, Wedell N (2008) **Selfish genetic elements and sexual selection: their impact on male fertility** *Genetica* **134**:99–111

Sandler L, Novitski E (1957) **Meiotic drive as an evolutionary force** *Am Nat* **91**:105–110

Sutter A, Lindholm AK (2015) **Detrimental effects of an autosomal selfish genetic element on sperm competitiveness in house mice** *Proc Biol Sci* **282**

Tsai IJ, Bensasson D, Burt A, Koufopanou V (2005) **Population genomics of the wild yeast *Saccharomyces paradoxus*: Quantifying the life cycle** *Proc Natl Acad Sci USA* **105**:4957–62

Tusso S, Suo F, Liang Y, Du LL, Wolf JBW (2022) **Reactivation of transposable elements following hybridization in fission yeast** *Genome Res* **32**:324–336

Zanders SE, Unckless RL (2019) **Fertility Costs of Meiotic Drivers** *Curr Biol* **29**:R512–R520

Zhang D, Gao F, Jakovlić I, Zou H, Zhang J, Li WX, Wang GT (2020) **PhyloSuite: An integrated and scalable desktop platform for streamlined molecular sequence data management and evolutionary phylogenetics studies** *Mol Ecol Resour* **20**:348–355

Zheng JX, Du TY, Shao GC, Ma ZH, Jiang ZD, Hu W, Suo F, He W, Dong MQ, Du LL (2023) **Ubiquitination-mediated Golgi-to-endosome sorting determines the toxin-antidote duality of fission yeast wtf meiotic drivers** *Nat Commun* **14**:1–8334

Zhou ZJ, Qiu Y, Pu Y, Huang X, Ge XY (2020) **BioAider: An efficient tool for viral genome analysis and its application in tracing SARS-CoV-2 transmission** *Sustain Cities Soc* **63**

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Reviewer #1 (Public review):**Summary**

The authors determine the phylogenetic relation of the roughly two dozen wtf elements of 21 *S. pombe* isolates and show that none of them in the original *S. pombe* are essential for robust mitotic growth. It would be interesting to test their meiotic function by simply crossing each deletion mutant with the parent and analyzing spores for non-Mendelian inheritance. If this has been reported already, that information should be added to the manuscript. If not, I suggest the authors do these simple experiments and add this information.

Strengths:

The most interesting data (Figure 4) show that one recombinant (wtfC4) between wtf18 and wtf23 produces in mitotic growth a poison counteracted by its own antidote but not by the parental antidotes. Again, it would be interesting to test this recombinant in a more natural setting - meiosis between it and each of the parents.

Weaknesses:

In the opinion of this reviewer, some minor rewriting is needed.

<https://doi.org/10.7554/eLife.100638.1.sa3>

Reviewer #2 (Public review):**Summary:**

This important study provides a mechanism that can explain the rapid diversification of poison-antidote pairs (wtf genes) in fission yeast: recombination between existing genes.

Strengths:

The authors analyzed the diversity of wtf in *S. pombe* strains, and found pervasive copy number variations. They further detected signals of recurrent recombination in wtf genes. To address whether recombination can generate novel wtf genes, the authors performed artificial recombination between existing wtf genes, and showed that indeed a new wtf can be generated: the poison cannot be detoxified by the antidotes encoded by parental wtf genes but can be detoxified by own antidote.

Weaknesses:

The study can benefit from demonstrating that the novel poison-antidote constructed by the authors can serve as a meiotic driver.

<https://doi.org/10.7554/eLife.100638.1.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript, Wang and colleagues explore factors contributing to the diversification of wtf meiotic drivers. wtf genes are autonomous, single-gene poison-antidote meiotic drivers that encode both a spore-killing poison (short isoform) and an antidote to the poison (long isoform) through alternative transcriptional initiation. There are dozens of wtf drivers present in the genomes of various yeast species, yet the evolutionary forces driving their diversification remain largely unknown. This manuscript is written in a straightforward and effective manner, and the analyses and experiments are easy to follow and interpret. While I find the research question interesting and the experiments persuasive, they do not provide any deeper mechanistic understanding of this gene family.

Strengths:

- (1) The authors present a comprehensive compendium and analysis of the evolutionary relationships among wtf genes across 21 strains of *S. pombe*.
- (2) The authors found that a synthetic chimeric wtf gene, combining exons 1-5 of wtf23 and exon 6 of wtf18, behaves like a meiotic driver that could only be rescued by the chimeric antidote but neither of the parental antidotes. This is a very interesting observation that could account for their inception and diversification.

Weaknesses:

(1) Deletion strains

The authors separately deleted all 25 Wtf genes in the *S. pombe* reference strain. Next, the authors performed a spot assay to evaluate the effect of wtf gene knockout on the yeast growth. They report no difference to the WT and conclude that the wtf genes might be largely neutral to the fitness of their carriers in the asexual life cycle at least in normal growth conditions.

The authors could have conducted additional quantitative growth assays in yeast, such as growth curves or competition assays, which would have allowed them to detect subtle fitness effects that cannot be quantified with a spot assay. Furthermore, the authors do not rule out

simpler explanations, such as genetic redundancy. This could have been addressed by crossing mutants of closely related paralogs or editing multiple wtf genes in the same genetic background.

Another concern is the lack of detailed information about the 25 knockout strains used in the study. There is no information provided on how these strains were generated or, more importantly, validated. Many of these wtf genes have close paralogs and are flanked by repetitive regions, which could complicate the generation of such deletion strains. As currently presented, these results would be difficult to replicate in other labs due to insufficient methodological details

(2) Lack of controls

The authors found that a synthetic chimeric wtf gene, constructed by combining exons 1-5 of wtf23 and exon 6 of wtf18, behaves as a meiotic driver that can be rescued only by its corresponding chimeric antidote, but not by either of the parental antidotes (Figure 4F). In contrast, three other chimeric wtf genes did not display this property (Figure 4C-E). No additional experiments were conducted to explain these differences, and basic control experiments, such as verifying the expression of the chimeric constructs, were not performed to rule out trivial explanations. This should be at the very least discussed. Also, it would have been better to test additional chimeras.

3. Statistical analyses

In line 130 the authors state that: "Given complex phylogenetic mixing observed among wtf genes (Figure 1E), we tested whether recombination occurred. We detected signals of recombination in the 25 wtf genes of the *S. pombe* reference genome ($p = 0$) and in the wtf genes of the 21 *S. pombe* strains ($p = 0$) using pairwise homoplasmy index (HPI) test. ". Reporting a p-value of 0 is not appropriate. Exact P-values should be reported.

<https://doi.org/10.7554/eLife.100638.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary

*The authors determine the phylogenetic relation of the roughly two dozen wtf elements of 21 *S. pombe* isolates and show that none of them in the original *S. pombe* are essential for robust mitotic growth. It would be interesting to test their meiotic function by simply crossing each deletion mutant with the parent and analyzing spores for non-Mendelian inheritance. If this has been reported already, that information should be added to the manuscript. If not, I suggest the authors do these simple experiments and add this information.*

Thanks for the great summary! Most of the wtf genes have been tested for meiotic drive phenotypes previously by Bravo Nunez et al. (2020; <http://doi.org/10.1371/journal.pgen.1008350>). The reference was cited in our original manuscript, and we added the details in the revised manuscript.

Strengths:

The most interesting data (Figure 4) show that one recombinant (wtfC4) between wtf18 and wtf23 produces in mitotic growth a poison counteracted by its own antidote but not by the parental antidotes. Again, it would be interesting to test this recombinant in a more natural setting - meiosis between it and each of the parents.

We will test the meiotic driver phenotype of the wtfC4 we constructed in *S. pombe* as suggested.

Weaknesses:

In the opinion of this reviewer, some minor rewriting is needed.

We did the rewriting as this reviewer suggested in the comments to authors.

Reviewer #2 (Public review):

Summary:

This important study provides a mechanism that can explain the rapid diversification of poison-antidote pairs (wtf genes) in fission yeast: recombination between existing genes.

Thanks!

Strengths:

The authors analyzed the diversity of wtf in S. pombe strains, and found pervasive copy number variations. They further detected signals of recurrent recombination in wtf genes. To address whether recombination can generate novel wtf genes, the authors performed artificial recombination between existing wtf genes, and showed that indeed a new wtf can be generated: the poison cannot be detoxified by the antidotes encoded by parental wtf genes but can be detoxified by own antidote.

Thanks for the great summary!

Weaknesses:

The study can benefit from demonstrating that the novel poison-antidote constructed by the authors can serve as a meiotic driver.

We will test the meiotic driver phenotype of the wtfC4 we constructed in *S. pombe* as suggested.

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In this manuscript, Wang and colleagues explore factors contributing to the diversification of wtf meiotic drivers. wtf genes are autonomous, single-gene poison-antidote meiotic drivers that encode both a spore-killing poison (short isoform) and an antidote to the poison (long isoform) through alternative transcriptional initiation. There are dozens of wtf drivers present in the genomes of various yeast species, yet the evolutionary forces driving their diversification remain largely unknown. This manuscript is written in a straightforward and effective manner, and the analyses and experiments are easy to follow and interpret. While I find the research question interesting and the experiments persuasive, they do not provide any deeper mechanistic understanding of this gene family.

Thanks! Please see the following for our point-to-point response.

Strengths:

*(1) The authors present a comprehensive compendium and analysis of the evolutionary relationships among wtf genes across 21 strains of *S. pombe*.*

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We will generate growth curves for all the 25 wtf deletion strains. We will also provide detailed for wtf gene knockout. However, for 25 wtf genes, there are too many combinations for editing two genes, and it is technically challenging to knock out multiple wtf together. Nevertheless, our results suggest single wtf gene has little effect on the host fitness under normal condition.

(2) Lack of controls

The authors found that a synthetic chimeric wtf gene, constructed by combining exons 1-5 of wtf23 and exon 6 of wtf18, behaves as a meiotic driver that can be rescued only by its corresponding chimeric antidote, but not by either of the parental antidotes (Figure 4F). In contrast, three other chimeric wtf genes did not display this property (Figure 4C-E). No additional experiments were conducted to explain these differences, and basic control experiments, such as verifying the expression of the chimeric constructs, were not performed to rule out trivial explanations. This should be at the very least discussed. Also, it would have been better to test additional chimeras.

We will verify the expression of the chimeric genes, and test the phenotype of meiotic diver for wtfC4 in *S. pombe*.

(3) Statistical analyses

In line 130 the authors state that: "Given complex phylogenetic mixing observed among wtf genes (Figure 1E), we tested whether recombination occurred. We detected signals of recombination in the 25 wtf genes of the S. pombe reference genome ($p = 0$) and in the wtf genes of the 21 S. pombe strains ($p = 0$) using pairwise homoplasy index (HPI) test. ". Reporting a p-value of 0 is not appropriate. Exact P-values should be reported.

We will report the exact p values in the revised manuscript.

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